A Technical Introduction to Feature selection

Section 1

m R M R = max S [1 | S | ∑ f i ∈ S I (f i ; c) − 1 | S | 2 ∑ f i , f j ∈ S I (f i ; f j)] . {\displaystyle \mathrm {mRMR} =\max _{S}\left[{\frac {1}{|S|}}\sum _{f_{i}\in S}I(f_{i};c)-{\frac {1}{|S|^{2}}}\sum _{f_{i},f_{j}\in S}I(f_{i};f_{j})\right].}

Section 2

Regularized trees The features from a decision tree or a tree ensemble are shown to be redundant. A recent method called regularized tree can be used for feature subset selection. Regularized trees penalize using a variable similar to the variables selected at previous tree nodes for splitting the current node. Regularized trees only need build one tree model (or one tree ensemble model) and thus are computationally efficient. Regularized trees naturally handle numerical and categorical features, interactions and nonlinearities. They are invariant to attribute scales (units) and insensitive to outliers, and thus, require little data preprocessing such as normalization. Regularized random forest (RRF) is one type of regularized trees. The guided RRF is an enhanced RRF which is guided by the importance scores from an ordinary random forest.

Section 3

m R M R = max x ∈ { 0 , 1 } n [∑ i = 1 n c i x i ∑ i = 1 n x i − ∑ i , j = 1 n a i j x i x j (∑ i = 1 n x i) 2] . {\displaystyle \mathrm {mRMR} =\max _{x\in \{0,1\}^{n}}\left[{\frac {\sum _{i=1}^{n}c_{i}x_{i}}{\sum _{i=1}^{n}x_{i}}}-{\frac {\sum _{i,j=1}^{n}a_{ij}x_{i}x_{j}}{(\sum _{i=1}^{n}x_{i})^{2}}}\right].}

Section 4

Calculate the mutual information as score for between all features (f i ∈ F {\displaystyle f_{i}\in F}) and the target class (c) Select the feature with the largest score (e.g. argmax f i ∈ F (I (f i , c)) {\displaystyle {\underset {f_{i}\in F}{\operatorname {argmax} }}(I(f_{i},c))}) and add it to the set of selected features (S) Calculate the score which might be derived from the mutual information Select the feature with the largest score and add it to the set of select features (e.g. argmax f i ∈ F (I d e r i v e d (f i , c)) {\displaystyle {\underset {f_{i}\in F}{\operatorname {argmax} }}(I_{derived}(f_{i},c))}) Repeat 3. and 4. until a certain number of features is selected (e.g. | S | = l {\displaystyle |S|=l}) The simplest approach uses the mutual information as the "derived" score. However, there are different approaches, that try to reduce the redundancy between features.

Section 5

where F n × 1 = [I (f 1 ; c) , … , I (f n ; c)] T {\displaystyle F_{n\times 1}=[I(f_{1};c),\ldots ,I(f_{n};c)]^{T}} is the vector of feature relevancy assuming there are n features in total, H n × n = [I (f i ; f j)] i , j = 1 … n {\displaystyle H_{n\times n}=[I(f_{i};f_{j})]_{i,j=1\ldots n}} is the matrix of feature pairwise redundancy, and x n × 1 {\displaystyle \mathbf {x} _{n\times 1}} represents relative feature weights. QPFS is solved via quadratic programming. It is recently shown that QFPS is biased towards features with smaller entropy, due to its placement of the feature self redundancy term I (f i ; f i) {\displaystyle I(f_{i};f_{i})} on the diagonal of H.

Section 6

Correlation feature selection The correlation feature selection (CFS) measure evaluates subsets of features on the basis of the following hypothesis: "Good feature subsets contain features highly correlated with the classification, yet uncorrelated to each other". The following equation gives the merit of a feature subset S consisting of k features:

Section 7

Embedded methods have been recently proposed that try to combine the advantages of both previous methods. A learning algorithm takes advantage of its own variable selection process and performs feature selection and classification simultaneously, such as the FRMT algorithm.

Section 8

Further reading Guyon, Isabelle; Elisseeff, Andre (2003). "An Introduction to Variable and Feature Selection". Journal of Machine Learning Research. 3: 1157–1182. Harrell, F. (2001). Regression Modeling Strategies. Springer. ISBN 0-387-95232-2. Liu, Huan; Motoda, Hiroshi (1998). Feature Selection for Knowledge Discovery and Data Mining. Springer. ISBN 0-7923-8198-X. Liu, Huan; Yu, Lei (2005). "Toward Integrating Feature Selection Algorithms for Classification and Clustering". IEEE Transactions on Knowledge and Data Engineering. 17 (4): 491–502. Bibcode:2005IDSO...17..491L. doi:10.1109/TKDE.2005.66. S2CID 1607600.

Section 9

Q P F S : min x { α x T H x − x T F } s.t. ∑ i = 1 n x i = 1 , x i ≥ 0 {\displaystyle \mathrm {QPFS} :\min _{\mathbf {x} }\left\{\alpha \mathbf {x} ^{T}H\mathbf {x} -\mathbf {x} ^{T}F\right\}\quad {\mbox{s.t.}}\ \sum _{i=1}^{n}x_{i}=1,x_{i}\geq 0}

Section 10

S P E C C M I : max x { x T Q x } s.t. ■ x ■ = 1 , x i ≥ 0 {\displaystyle \mathrm {SPEC_{CMI}} :\max _{\mathbf {x} }\left\{\mathbf {x} ^{T}Q\mathbf {x} \right\}\quad {\mbox{s.t.}}\ \|\mathbf {x} \|=1,x_{i}\geq 0}

Section 11

Exhaustive Best first Simulated annealing Genetic algorithm Greedy forward selection Greedy backward elimination Particle swarm optimization Targeted projection pursuit Scatter search Variable neighborhood search Two popular filter metrics for classification problems are correlation and mutual information, although neither are true metrics or 'distance measures' in the mathematical sense, since they fail to obey the triangle inequality and thus do not compute any actual 'distance' – they should rather be regarded as 'scores'. These scores are computed between a candidate feature (or set of features) and the desired output category. There are, however, true metrics that are a simple function of the mutual information; see here. Other available filter metrics include:

Section 12

C F S = max S k [r c f 1 + r c f 2 + ■ + r c f k k + 2 (r f 1 f 2 + ■ + r f i f j + ■ + r f k f k − 1)] . {\displaystyle \mathrm {CFS} =\max _{S_{k}}\left[{\frac {r_{cf_{1}}+r_{cf_{2}}+\cdots +r_{cf_{k}}}{\sqrt {k+2(r_{f_{1}f_{2}}+\cdots +r_{f_{i}f_{j}}+\cdots +r_{f_{k}f_{k-1}})}}}\right].}

Section 13

H S I C L a s s o : min x 1 2 ∑ k , l = 1 n x k x l HSIC (f k , f l) − ∑ k = 1 n x k HSIC (f k , c) + λ ■ x ■ 1 , s.t. x 1 , … , x n ≥ 0 , {\displaystyle \mathrm {HSIC_{Lasso}} :\min _{\mathbf {x} }{\frac {1}{2}}\sum _{k,l=1}^{n}x_{k}x_{l}{\mbox{HSIC}}(f_{k},f_{l})-\sum _{k=1}^{n}x_{k}{\mbox{HSIC}}(f_{k},c)+\lambda \|\mathbf {x} \|_{1},\quad {\mbox{s.t.}}\ x_{1},\ldots ,x_{n}\geq 0,}